

# VELAVAN R

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🌐 <https://github.com/velavanzoro>    📁 Portfolio

## Education

**B.E – Computer Science & Engineering (AI & ML),** **09/2023 - 05/2027**  
Chennai Institute Of Technology  
Achieved a CGPA of 7.5/10, reflecting consistent academic performance and mastery of core coursework.

## Experience

**Full Stack Development Intern — Gowri Software Solutions Pvt. Ltd, Bangalore** **05/2025 - 06/2025**  
• Built and integrated front-end and back-end modules using React, Node.js, Express, improving feature delivery speed by 30%. Implemented REST APIs, optimized database queries, and enhanced UI responsiveness, reducing page load time by 25%.  
• Collaborated within an Agile team to deliver 12+ production-ready full-stack components with 100% on-time sprint completion, contributing to a 30% increase in feature delivery efficiency.

**Research & Development Intern — Chennai Institute of Technology** **10/2024 - 12/2024**  
• Conducted research on RF-based drone detection, analyzing datasets and benchmarking deep learning models for classification accuracy.  
• Compiled a comprehensive review of scalable detection methods and improved model sensitivity by 12% using optimized preprocessing.

## Projects

**TB Detection from Chest X-rays using Deep Learning** **08/2025 - 10/2025**  
• Engineered a CNN-based medical imaging system achieving 94% accuracy using ResNet/DenseNet with transfer learning.  
• Enhanced sensitivity by 13% and reduced false negatives through hyperparameter tuning and augmentation strategies.  
• Built an end-to-end pipeline covering data cleaning, training, evaluation, and visualization.  
• Technologies utilized: PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Google Colab

**Interactive Storytelling with AI-Generated Images** **06/2025 - 08/2025**  
• Built an interactive AI storytelling platform using React + FastAPI, enabling real-time story generation.  
• Integrated LLaMA2 + Hugging Face Diffusers, achieving 3–5 sec/image generation speed with 90% content coherence.  
• Improved API response flow by 20% using optimized model loading and cache management.  
• Technologies utilized: React, FastAPI, LLaMA2, Diffusers, Python

**License Plate Recognition System** **11/2024 - 12/2024**  
• Engineered an OCR-based LPR model with 93% recognition accuracy for images and video streams.  
• Automated license plate segmentation and tracking, reducing manual verification time by 60%.  
• Deployed model for test scenarios involving traffic automation and vehicle identification.  
• Utilized OpenCV, Tesseract OCR, Python, and NumPy to support project objectives.

## Technologies

|                                                                                   |                                                                      |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Programming Languages:</b><br>Python, Java, C++, JavaScript, SQL               | <b>Data Analytics:</b><br>Excel, Power BI, Pandas, NumPy, Statistics |
| <b>AI &amp; Machine Learning:</b><br>TensorFlow, PyTorch, CNN, Deep Learning, NLP | <b>Web Development:</b><br>React.js, Node.js, Express.js, FastAPI    |
| <b>Databases:</b><br>MySQL, MongoDB                                               | <b>Tools:</b><br>Git, GitHub, VS Code, Colab                         |

## Certifications

- |                                                                      |                                                                 |                                               |
|----------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------|
| • Cisco Networking Academy – Introduction to Cybersecurity, CyberOps | • Coursera-Data Structures & Algorithms Specialization          | • Simplilearn- SQL Basics, Java Servlet & JSP |
| • EduSkills-AWS IoT Cloud Engineer Internship                        | • NPTEL-IOT, Data Analytics With Python, Reinforcement Learning |                                               |

## Competitive Programming

- **LeetCode:** Max Rating 1814, 600+ problems solved, Top 30% globally, Badges: 5
- **CodeChef:** Max Rating 1377, Division 4 Highest Rank: 5143, Contest Rank: 6402
- Strengthened algorithmic thinking and efficiently solved complex coding challenges under time constraints.

## Volunteering

- Conducted workshops on Git & GitHub, teaching version control and open-source collaboration.
- Hospitality Lead for college symposium; coordinated 200+ participants, ensuring smooth event execution.
- Mentored juniors in Python basics and competitive programming.
- Supported logistics and execution for multiple technical events and workshops.

## Achievements

- Reached final round of Ideathon competition.
- Qualified GATE 2026 (Data Science & Artificial Intelligence) with a score of 405 and All India Rank (AIR) 6110.